Embedded System for Automatic Plant Irrigation based on micro climate sensor

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***Abstract*— This paper presents the development of an embedded system for automatic plant irrigation designed for domestic environments. The system integrates a capacitive soil moisture sensor, two thermistors for microclimate measurement, and a water level sensor, all connected to a microcontroller that performs real-time decision-making and controls a 5V submersible water pump through a 5V relay module. By combining soil moisture and leaf temperature data, the system determines irrigation needs more precisely, improving water efficiency and plant health.**

Preliminary qualitative tests with a physical prototype demonstrated correct sensor readings, reliable control logic, and effective actuation, validating the feasibility of a low-cost, autonomous, and sustainable irrigation solution. The system accomplishes its main goal by keeping plants alive with adjusted parameters for specific plants

***Keywords*— Soil Moisture Monitoring, Automatic Irrigation, Water Management, Leaf Microclimate Sensor**

I. INTRODUCTION

In urban environments and modern households, maintaining healthy indoor and outdoor plants can be a challenge due to busy schedules and inconsistent care. Traditional manual watering practices often lead to common problems such as over- or under-watering, which result in suboptimal plant health and water waste. In this context, smart irrigation systems based on the Internet of Things (IoT) have emerged as innovative solutions to automate and optimize plant care in domestic settings [1], [2]. These systems integrate sensors and connected devices to collect real-time data, supporting automated decisions that improve both water efficiency and plant vitality.

While existing IoT-based irrigation systems have demonstrated significant potential, many focus primarily on soil moisture or water level monitoring [3], overlooking other critical variables that directly influence plant health. In particular, the microclimate at the leaf surface provides a more accurate reflection of the plant's physiological condition, as it captures the interaction between environmental variables and the crop itself [4]. To address this gap, this work presents the design and implementation of an embedded system for automatic plant irrigation that integrates soil moisture monitoring[5], reservoir water level detection [6], and leaf microclimate sensing [2] for domestic environments. By combining data from both soil and foliage, the proposed approach enhances irrigation decision-making, reduces water waste, and supports more precise and sustainable agricultural practices.

The rest of this work is organized as follows. Section II reviews related studies on IoT-based irrigation and plant monitoring systems. Section III details the design and architecture of the proposed embedded system, including hardware and sensor integration. Section IV presents and discusses the system's performance through experimental evaluation. Finally, Section V concludes the work and outlines potential directions for future improvements.

II. RELATED WORK

Recent research on smart irrigation and precision agriculture has focused on integrating low-cost sensors and embedded systems to optimize water usage and automate plant care. The majority of these works rely on the Internet of Things (IoT) [7] paradigm, where distributed sensor nodes collect environmental data to support autonomous irrigation decisions [1].

Several studies have explored soil moisture-based irrigation systems [5] using microcontrollers such as Arduino or ESP32, which allow simple calibration and control of water pumps through threshold-based algorithms. These approaches demonstrate the feasibility of affordable irrigation automation [8] but often neglect additional environmental variables that can affect plant health, such as humidity, temperature, and leaf transpiration [2].

Other works have proposed IoT architectures that combine soil and climatic data with wireless communication, enabling remote monitoring and control via mobile or web interfaces [9]. These systems typically use Wi-Fi or GSM connectivity to transmit data to a cloud platform, where irrigation can be triggered automatically or manually by the user. However, these implementations depend heavily on network stability and are often designed for larger-scale agricultural environments.

Recent advances in agricultural sensing have emphasized the importance of incorporating plant-level measurements. For instance, [4] introduced a microclimate sensor capable of measuring temperature and humidity directly at the leaf surface, providing more precise insights into plant physiological conditions [10]. Such data can significantly improve irrigation accuracy by correlating environmental parameters with the plant's actual water demand.

In contrast to previous works that focus primarily on soil or atmospheric conditions, the proposed system combines soil moisture, reservoir level, and leaf microclimate measurements in a single embedded platform. This integration enables context-aware irrigation decisions and local actuation without relying on external cloud processing.

III. THE METHOD

The developed system operates through a sequence of stages that ensure autonomous and efficient irrigation control based

on real-time environmental data. Its operation can be divided into four main phases: data acquisition, decision processing, actuation, and user feedback. Figure 1 illustrates the logical flow of these processes through an activity diagram.

A. Data Acquisition

The system begins its operation by continuously collecting sensor data from three main sources: the capacitive soil moisture sensor, two thermistors (one for ambient temperature and one placed close to the leaf surface), and the ultrasonic water level sensor. The soil sensor measures the volumetric water content of the substrate, providing an indicator of whether the plant requires watering. In parallel, the ultrasonic sensor monitors the reservoir water level, preventing the pump from operating when water is insufficient. The thermistors capture microclimate temperature conditions that influence the irrigation decision.

All sensors are periodically read by the Arduino Mega using analog or digital input pins. To reduce electrical noise and improve stability, each measurement undergoes a filtering process based on the moving average of the most recent samples. These readings are then normalized and stored for later processing.

B. Decision Processing

After data acquisition, the microcontroller executes the irrigation control algorithm. This algorithm compares the soil moisture reading against a predefined threshold representing the optimal level for the plant species. If the value is below this limit, the system evaluates the temperature difference between the two thermistors. A high leaf temperature combined with low humidity suggests water deficit and triggers irrigation.

Conversely, if the ambient temperature is low or soil humidity remains high, the system postpones activation to avoid over watering. This decision process ensures that irrigation occurs only when necessary, using both soil and leaf conditions to improve precision.

Rigel: legal! qual seria o tempo de espera?

C. Actuation

When irrigation is required, the microcontroller activates a 5V submersible water pump through a 5V relay module. The relay isolates the control circuit from the power circuit, ensuring safe switching. The pump remains active until the soil moisture sensor detects a satisfactory level or a timeout condition is reached. During the operation, the system continuously monitors the water level to prevent dry running and hardware damage.

D. User Feedback

Three LEDs provide visual feedback: a green LED indicates that irrigation is active, a red LED signals a low or empty water reservoir and a yellow LED indicates system stand by. In future versions, wireless communication (via Wi-Fi or Bluetooth) will allow user notifications to be sent to mobile devices.

E. System Behavior Summary

The overall logic of the system is summarized in the activity diagram shown in Figure 1. Initially, sensors measure soil moisture, reservoir level, and leaf temperature. If the tank is empty, the system halts irrigation and alerts the user. Otherwise, it processes the data and determines whether irrigation should begin. When irrigation is necessary, the pump is activated; otherwise, the system waits before repeating the cycle.

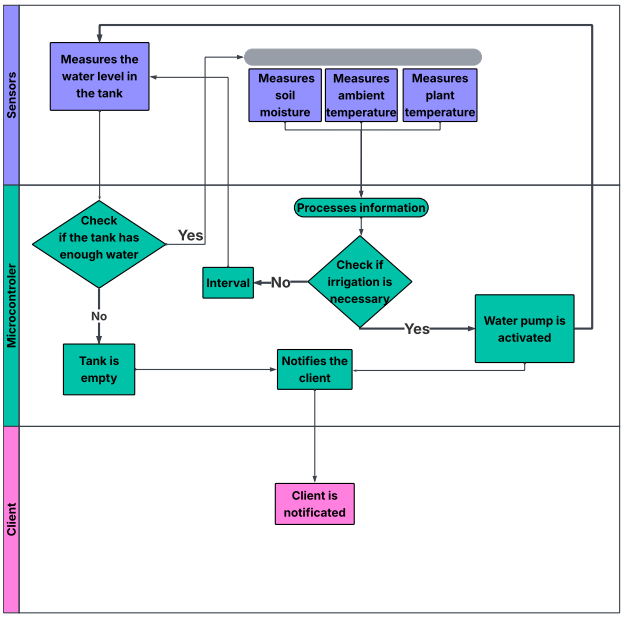

Fig. 1. Activity diagram of the irrigation process.

F. Prototype

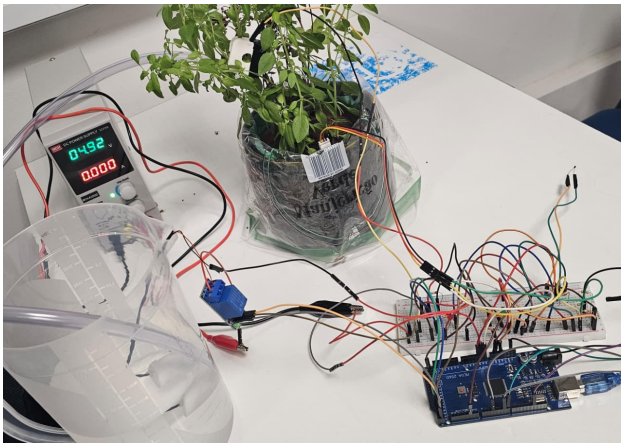

Fig. 2. Prototype of the automatic irrigation system assembled on a protoboard.

As shown in Figure 2, the physical prototype was built using an Arduino Mega, a capacitive soil moisture sensor, two NTC thermistors, and an HC-SR04 ultrasonic sensor. A 5V submersible water pump connected through a 5V relay module controls irrigation. The system also includes three LEDs: a green one for irrigation activation, a red one for

reservoir warnings and a yellow one for system stand by. All components are powered by a 5V supply, forming a compact embedded platform for testing and validation.

IV. Experimental Results

The prototype was assembled and tested to verify the correct operation of the sensors, control logic, and actuation mechanisms. All components including the capacitive soil moisture sensor, two NTC thermistors, the HC-SR04 ultrasonic level sensor, and a 5V submersible pump controlled via a 5V relay module were connected to the Arduino Mega and powered by a 5V supply.

A. Experimental Setup

The system was tested in a controlled environment simulating domestic plant care conditions. The capacitive soil sensor was inserted into a small soil sample to emulate real moisture variations, while the ultrasonic sensor measured the distance to the water surface in a reservoir. The two thermistors were positioned near a leaf and in ambient air to detect microclimate changes. Three LEDs provided status feedback: the green LED indicated active irrigation, the yellow LED signaled that the system was in standby mode with sufficient water in the reservoir, and the red LED warned of a low or empty water level. [Figure 2]

B. Qualitative Observations

Prototype testing involved simulating different environmental conditions by manually varying the sensor inputs. In all cases, the system behaved as expected. When the water level was low (HC-SR04 $>$ 25 cm), the red LED turned on and the pump remained off, ensuring protection. Under normal and standby conditions, the yellow or green LEDs indicated appropriate states, with the pump activating only when soil humidity dropped (ADC $>$ 301.5), although the soil moisture sensor had troubles reading humidity that's on the other side of the vase. Temperature variations correctly adjusted irrigation timing: early under hot conditions, delayed under cold ones, and immediate when leaf temperature exceeded ambient by more than 2°C. These results confirm that the control logic and sensing strategy operated reliably throughout all scenarios.

The results confirmed that the system responded correctly to all test conditions. The relay reliably switched the submersible pump, and the protection logic prevented actuation during low-water conditions. Temperature variations detected by the thermistors influenced irrigation timing as expected, validating the contribution of the leaf microclimate measurement.

1) System Behavior Evaluation: During continuous operation, the system demonstrated stable behavior and responsiveness to sensor variations. The delay between condition detection and actuation was below one second, ensuring near real-time control suitable for small-scale domestic irrigation. The LEDs provided clear and intuitive feedback, allowing easy monitoring of the irrigation state without external interfaces.

The integration of microclimate sensing proved beneficial: when the leaf thermistor indicated elevated temperature and the soil moisture was low, the system triggered irrigation earlier, while under cooler or humid conditions, irrigation was delayed or suspended. This adaptive response confirmed the system's ability to combine multiple environmental variables for more efficient water management.

2) Discussion: These qualitative results validate the correct integration and behavior of all hardware components. The capacitive soil moisture sensor offered reliable readings even though it had a small radius, while the dual thermistor setup improved precision in assessing plant conditions. The 5V relay-controlled submersible pump ensured safe and autonomous operation, and the LED indicators enhanced user awareness of system status.

While this stage focused on qualitative validation, future work will include quantitative analyses of sensor calibration and radius, irrigation duration, power consumption, and water usage efficiency to fully characterize the system's performance and sustainability.

V. Conclusion

This work presented the design and implementation of an embedded system for automatic plant irrigation that integrates a capacitive soil moisture sensor, two thermistors for microclimate and ambient temperature measurement, and a submersible 5V water pump controlled by a 5V relay module. The system was successfully tested in a controlled environment, demonstrating reliable autonomous operation and real-time feedback to the user through LED indicators.

Experimental observations confirmed that the system accurately responds to varying soil moisture, temperature, and water level conditions. The intelligent decision algorithm proved effective in adjusting irrigation behavior based on environmental conditions: it prioritized irrigation under high ambient temperature or plant heat stress, and delayed it in cooler environments, thus avoiding unnecessary water consumption.

By combining soil and leaf-level sensing with simple embedded control, the proposed prototype promotes efficient and sustainable water management for domestic plant care. Future work includes integrating wireless connectivity for remote monitoring using MQTT protocol, developing a mobile notification interface, and conducting long-term performance tests with different plant species to further validate the decision thresholds.

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Rigel: o nome dos autores deveria ser o nome completo, podemos abreviar os primeiro sobrenomes mas deixar o nome completo é importante para a catalogação desses trabalhos